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Quentin Rebergue - Graduate Software Engineer - Systems & Virtualization

Profile

Looking for a full-time position in software engineering with a focus on ystems, Virtualization, Low-Level, C++.

MSc in Computer Engineering (EPITA - GISTRE) in Systems / Embedded.

Fluent in French and English (TOEIC 915, C1). Interests: watchmaking, music, hiking/running , cooking.

Work Experience

- **Software Engineer Intern - Platform team, ARISTA**, Dublin 02/2026 – 08/2026

  Implemented non-zero reset value management for ASIC Physical Table registers across the chip-agnostic Strata infrastructure (SDK parsing, code generation, DMA) in python and C++. Won 1st place at Arista's Hackathon by building an AI agent optimization tool (70% token reduction on internal codebases). Approved for production development.

- **Head of Teaching Assistants - YAKA** / **ACU Program, EPITA**, Kremlin-Bicetre 09/2024 – 02/2026

  Led the national team of 67 TAs and supervised 5 intensive coding bootcamps (Java, C++, JS, C/Bash, SQL) and 7 large-scale projects (compiler, HTTP server in C, malloc, POSIX shell) for 700+ 3rd-year students across 5 campuses. Implemented 1-on-1 code review mentorship model, improving TA-to-student ratio from 1:30 to 1:20.

School Projects

- **My Docker** – Built a lightweight container runtime in C, using Linux namespaces, pivot_root, capabilities dropping, seccomp syscall filtering, and cgroups. Pulls and mounts a root filesystem from a container registry.
- **RFID Driver** – Developed a Linux kernel character device driver for the MFRC522 RFID reader with file_operations callbacks, SPI regmap communication, and a FIFO command interface. Tested via QEMU emulation.
- **Linux System Tracing & Storage** – Used strace fault injection and syscall filtering to reverse-engineer a binary. Wrote a Bash script automating Linux storage stacks: loop devices, LUKS2, LVM, dm-linear, dm-delay, and RAID.
- **Unicycle CPU** – Designed a 32-bit single-cycle ARM-like processor in VHDL (ALU, register file, instruction decoder, control unit, conditional branching), synthesized on a real FPGA with Intel Quartus, and validated with testbenches.

Education

- **MSc Computer Engineering, Low-level & Systems (GISTRE)**, EPITA 2020 – 2026
- **Preparatory Class (PTSI) in Engineering Science**, Lycée Jules Ferry Versailles 2019 – 2020

Technical Expertise

- **Programming & Languages:** ASM, Bash, C, C++, C++20, CUDA, Rust, Shell.
- **Embedded & Hardware:** ASIC interfacing, FPGA, I2C, SPI, UART.
- **Other** / **DevOps:** GDB, Git, Kubernetes, Security, monitoring, optimisation, testing, valgrind.
- **Systems:** Linux, Kernel, Drivers, Bootloader, Virtualization, Multithreading, GPU, real-time, DMA, operational systems, low-latency.